CMSC 202 Spring 2026

Project 3 – Hospital Triage

Assignment: Project 3 – Hospital Triage

Due Date: Tuesday, March 31st @ 8:59pm on GL

Value: 80 points

1. Overview

In this project you will:

- Implement a linked-list data structure,
- Use dynamic memory allocation to create new objects,
- Practice using C++ class syntax,
- Practice object-oriented thinking.

2. Background

Hospitals rely on digital systems to manage emergency room (ER) triage operations, including patient intake, treatment assignment, and billing. These systems store information about available treatments such as procedure names, costs, and unique identification numbers, and dynamically track each patient's treatment record. In this project, you will build a simplified version of an ER triage system that loads a treatment catalog from a file, allows users to assign or remove treatments for a patient, and automatically computes subtotals, facility fees, and totals.

The program will use a linked list where each node represents one assigned treatment. This project provides practice with object-oriented design, linked list manipulation, and memory management in C++. Similar systems are used in real-world hospitals and emergency departments to track patient treatments and manage billing efficiently.

3. Assignment Description

Your assignment is to build a treatment catalog, patient record, and triage system that allows the user to simulate an ER. This system can only handle one patient at a time.

1. You will be coding seven files: **`Treatment.cpp`** (easy), **`TreatmentCatalog.cpp`** (medium), **`Patient.cpp`** (medium) ,

`PatientTreatment.cpp` (medium) **`TreatmentList.cpp`** (hard), **`and TriageSystem.cpp`** (hard but partially provided).

2. There is one input file. The input file, **`treatments.txt`**, lists all available treatments. For this project, each line in the input file will represent one treatment that can be administered to a patient (stored in **`TreatmentCatalog's m_treatments`**)

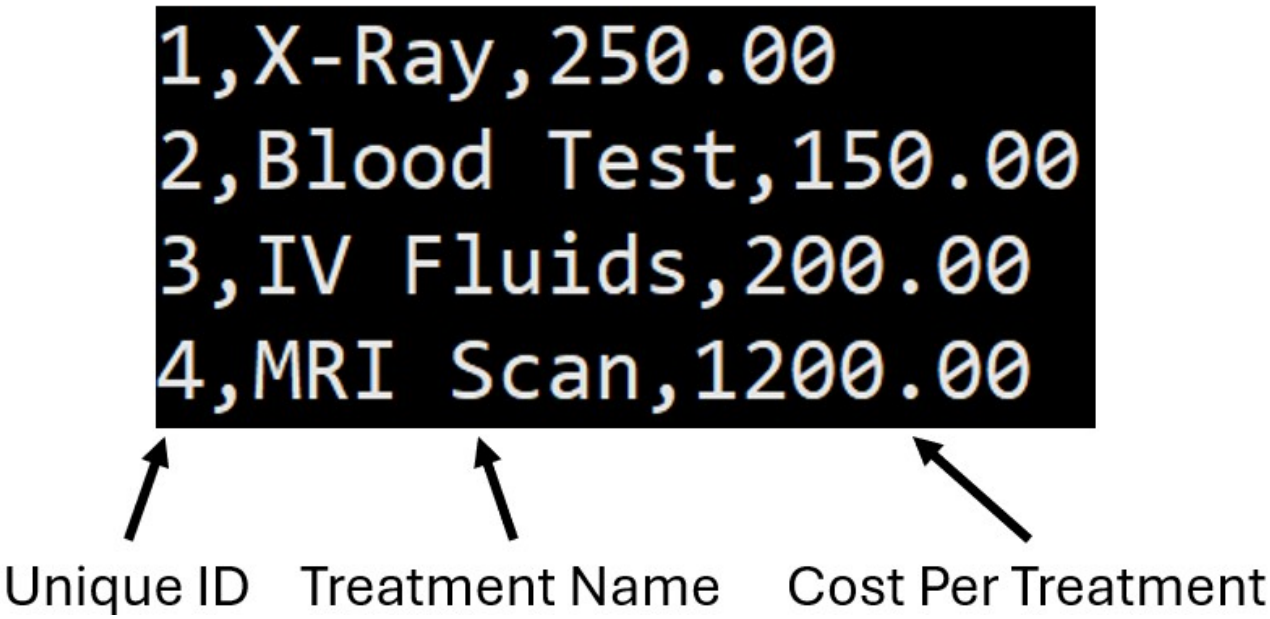

Figure 1. Input File

Treatment	Treatment available for patient (id, name, and cost per treatment)
TreatmentCatalog	List of all available Treatments (loaded from input file)
Patient	Contains an TreatmentList, computes subtotals, facility fee, and totals.
PatientTreatment	Holds one line of a treatment (1 x-ray, for example)
TreatmentList/Node	Implements a linked list to store order treatments.
TriageSystem	User interface class that loads the menu, accepts input, adds/removes treatments, prints record, and handles discharge.
proj3	Driver that contains main that creates a TriageSystem and runs it

Figure 2. Description of Classes/Files

3. The **`Node.h`** file is a node class that holds data for the patient record. It is used by the **`TreatmentList`** (which is the linked list).
4. The **`Treatments`** are the individual procedures that can be administered to a patient such as an X-Ray or Blood Test. **`TreatmentCatalog`** is made up of all of the available **`Treatments`**.
5. The **`PatientTreatment`** class holds one line of a patient record and **`Patient`** contains a **`TreatmentList`**.

4. Requirements:

Initially, you will have to use the following files **`Treatment.h`**, **`TreatmentCatalog.h`**, **`Patient.h`**, **`PatientTreatment.h`**, **`TreatmentList.h`**, **`TriageSystem.h`**, **`makefile`**, **`proj3.cpp`**, and one input files (**`treatments.txt`**). Additionally, there is one sample run available to view (**`proj3_sample1.txt`**). You can copy the files from Prof. Dixon's folder at:

`/afs/umbc.edu/users/j/d/jdixon/pub/cs202/proj3`

To copy it into your project folder, just navigate to your project 3 folder in your home folder and use the command:

```
cp /afs/umbc.edu/users/j/d/jdixon/pub/cs202/proj3/* .
```

Notice the trailing period is required (it says copy it into this folder).

- You must follow the coding standard as defined in the CMSC 202 coding standards (found on Blackboard under course policies). This includes comments as required.
- The project must be turned in on time by the deadline listed above.
- The project must be completed in C++. You may not use any libraries or data structures that we have not learned in class. Libraries we have learned include **`<iostream>`**, **`<fstream>`**, **`<iomanip>`**, **`<vector>`**, **`<cstdlib>`**, **`<time.h>`**, **`<cmath>`** and **`<string>`**. You should only use **`namespace std`**.

- Initially, you will have to use the following files **`Treatment.h`**, **`TreatmentCatalog.h`**, **`Patient.h`**, **`TreatmentList.h`**, **`PatientTreatment.h`**, **`TriageSystem.h`**, **`makefile`**, **`proj3.cpp`**, and one input file (**`treatments.txt`**). Do not edit the header files. Do not add functions to the header or cpp files. You need to code the .cpp files.
- All user inputs will be assumed to be the correct data type. For example, if you ask the user for an integer, they will provide an integer.
- Regardless of the sample output below, all user input must be validated. If you ask for a number between 1 and 5 with the user entering an 8, the user should be re-prompted.
- These are the specific restrictions for this project:
  - No recursion. No global variables. No for-each loops. No auto.
  - You may not use any functions, libraries, or data structures that we have not learned in class.
  - [redacted]
  - Do not use EOF to load the file but rather use a while loop and multiple getline commands. The input file can have any length. Here is an example: https://www.digitalocean.com/community/tutorials/getline-in-c-plus-plus Do not use stringstream. Hint: What are the data parts of each treatment? What is the treatment? You are allowed to use **`stoi`** to convert strings to integers and **`stod`** to convert strings to doubles. Here is an example: https://geeksprogramming.com/stoi-function/
  - No warnings, errors, or memory leaks when you turn it in.

5. Sample Input and Output

5.1. Sample Run

A normal run of the compiled code would look like this with user input highlighted in blue:

```
./proj3

== ER TRIAGE SYSTEM ==
1) View treatment catalog
2) Assign treatment to patient
3) Remove treatment from patient
4) View patient record
5) Discharge patient
0) Exit
Choice: 1
ID   Treatment                        Cost
-----------------------------------------
1    X-Ray                     $     250.00
2    Blood Test                $     150.00
3    IV Fluids                 $     200.00
4    MRI Scan                  $    1200.00
5    CT Scan                   $     800.00
6    EKG                       $     350.00
7    Urinalysis                $      75.00
8    Wound Closure             $     500.00
9    Pain Medication           $      50.00
10   Oxygen Therapy            $     175.00

== ER TRIAGE SYSTEM ==
1) View treatment catalog
2) Assign treatment to patient
3) Remove treatment from patient
4) View patient record
5) Discharge patient
0) Exit
Choice: 2
ID   Treatment                        Cost
```

```
-----------------------------------------
1    X-Ray                    $     250.00
2    Blood Test               $     150.00
3    IV Fluids                $     200.00
4    MRI Scan                 $    1200.00
5    CT Scan                  $     800.00
6    EKG                      $     350.00
7    Urinalysis               $      75.00
8    Wound Closure            $     500.00
9    Pain Medication          $      50.00
10   Oxygen Therapy           $     175.00
Enter treatment ID: 1
Enter number of doses/units: 1
Treatment assigned.

== ER TRIAGE SYSTEM ==
1) View treatment catalog
2) Assign treatment to patient
3) Remove treatment from patient
4) View patient record
5) Discharge patient
0) Exit
Choice: 3

== PATIENT RECORD #2001 ==
#   Treatment                 Doses   Unit Cost       Total
-------------------------------------------------------------
1   X-Ray                         1      250.00      250.00
-------------------------------------------------------------
                           Subtotal:                  250.00
                 Facility Fee (15%):                   37.50
                              Total:                  287.50
```

```
Remove which treatment #? 1
Removed treatment 1.

== ER TRIAGE SYSTEM ==
1) View treatment catalog
2) Assign treatment to patient
3) Remove treatment from patient
4) View patient record
5) Discharge patient
0) Exit
Choice: 4

== PATIENT RECORD #2001 ==
(no treatments assigned)

== ER TRIAGE SYSTEM ==
1) View treatment catalog
2) Assign treatment to patient
3) Remove treatment from patient
4) View patient record
5) Discharge patient
0) Exit
Choice: 2
ID   Treatment                          Cost
-----------------------------------------
1    X-Ray                     $     250.00
2    Blood Test                $     150.00
3    IV Fluids                 $     200.00
4    MRI Scan                  $    1200.00
5    CT Scan                   $     800.00
6    EKG                       $     350.00
7    Urinalysis                $      75.00
```

```
8    Wound Closure            $     500.00
9    Pain Medication          $      50.00
10   Oxygen Therapy           $     175.00
Enter treatment ID: 2
Enter number of doses/units: 3
Treatment assigned.

== ER TRIAGE SYSTEM ==
1) View treatment catalog
2) Assign treatment to patient
3) Remove treatment from patient
4) View patient record
5) Discharge patient
0) Exit
Choice: 4

== PATIENT RECORD #2001 ==
#   Treatment                 Doses   Unit Cost       Total
------------------------------------------------------------
1   Blood Test                    3      150.00      450.00
------------------------------------------------------------
                           Subtotal:                 450.00
                 Facility Fee (15%):                  67.50
                              Total:                 517.50

== ER TRIAGE SYSTEM ==
1) View treatment catalog
2) Assign treatment to patient
3) Remove treatment from patient
4) View patient record
5) Discharge patient
0) Exit
```

```
Choice: 0
```

Here is an example with some input validation.

```
./proj3

== ER TRIAGE SYSTEM ==
1) View treatment catalog
2) Assign treatment to patient
3) Remove treatment from patient
4) View patient record
5) Discharge patient
0) Exit
Choice: -1
Unknown choice.

== ER TRIAGE SYSTEM ==
1) View treatment catalog
2) Assign treatment to patient
3) Remove treatment from patient
4) View patient record
5) Discharge patient
0) Exit
Choice: 6
Unknown choice.

== ER TRIAGE SYSTEM ==
1) View treatment catalog
2) Assign treatment to patient
3) Remove treatment from patient
4) View patient record
5) Discharge patient
0) Exit
Choice: 2
ID   Treatment                        Cost
----------------------------------------
1    X-Ray                     $     250.00
```

```
2    Blood Test                $     150.00
3    IV Fluids                 $     200.00
4    MRI Scan                  $    1200.00
5    CT Scan                   $     800.00
6    EKG                       $     350.00
7    Urinalysis                $      75.00
8    Wound Closure             $     500.00
9    Pain Medication           $      50.00
10   Oxygen Therapy            $     175.00
Enter treatment ID: 11
Enter number of doses/units: 2
Invalid treatment ID or dose count.

== ER TRIAGE SYSTEM ==
1) View treatment catalog
2) Assign treatment to patient
3) Remove treatment from patient
4) View patient record
5) Discharge patient
0) Exit
Choice: 2
ID   Treatment                        Cost
----------------------------------------
1    X-Ray                     $     250.00
2    Blood Test                $     150.00
3    IV Fluids                 $     200.00
4    MRI Scan                  $    1200.00
5    CT Scan                   $     800.00
6    EKG                       $     350.00
7    Urinalysis                $      75.00
8    Wound Closure             $     500.00
9    Pain Medication           $      50.00
10   Oxygen Therapy            $     175.00
Enter treatment ID: 10
Enter number of doses/units: 2
Treatment assigned.
```

```
== ER TRIAGE SYSTEM ==
1) View treatment catalog
2) Assign treatment to patient
3) Remove treatment from patient
4) View patient record
5) Discharge patient
0) Exit
Choice: 3

== PATIENT RECORD #2001 ==
#   Treatment                 Doses   Unit Cost       Total
------------------------------------------------------------
1   Oxygen Therapy                2      175.00      350.00
------------------------------------------------------------
                          Subtotal:                  350.00
                Facility Fee (15%):                   52.50
                             Total:                  402.50
Remove which treatment #? 2
Invalid treatment number.

== ER TRIAGE SYSTEM ==
1) View treatment catalog
2) Assign treatment to patient
3) Remove treatment from patient
4) View patient record
5) Discharge patient
0) Exit
Choice: 3

== PATIENT RECORD #2001 ==
#   Treatment                 Doses   Unit Cost       Total
------------------------------------------------------------
1   Oxygen Therapy                2      175.00      350.00
------------------------------------------------------------
                          Subtotal:                  350.00
```

```
                Facility Fee (15%):                    52.50
                             Total:                   402.50
Remove which treatment #? 0
Invalid treatment number.

== ER TRIAGE SYSTEM ==
1) View treatment catalog
2) Assign treatment to patient
3) Remove treatment from patient
4) View patient record
5) Discharge patient
0) Exit
Choice: 4

== PATIENT RECORD #2001 ==
#   Treatment                 Doses   Unit Cost       Total
------------------------------------------------------------
1   Oxygen Therapy                2      175.00      350.00
------------------------------------------------------------
                           Subtotal:                  350.00
                Facility Fee (15%):                    52.50
                             Total:                   402.50

== ER TRIAGE SYSTEM ==
1) View treatment catalog
2) Assign treatment to patient
3) Remove treatment from patient
4) View patient record
5) Discharge patient
0) Exit
Choice: 5

== PATIENT RECORD #2001 ==
#   Treatment                 Doses   Unit Cost       Total
------------------------------------------------------------
1   Oxygen Therapy                2      175.00      350.00
```

```
-----------------------------------------------------------
                          Subtotal:                  350.00
                Facility Fee (15%):                   52.50
                             Total:                  402.50


*** PATIENT DISCHARGED ***


== ER TRIAGE SYSTEM ==
1) View treatment catalog
2) Assign treatment to patient
3) Remove treatment from patient
4) View patient record
5) Discharge patient
0) Exit
Choice: 0
Thank you for using Hospital Triage
```

There is one additional sample file with a longer run called: **`proj3_sample1.txt`**.

6. Compiling and Running

Because we are using a significant amount of dynamic memory for this project, you are required to manage any memory leaks that might be created. For a linked list, this is most commonly related to the dynamically allocated nodes. Remember, in general, for each item that is dynamically created, it should be deleted using a destructor.

One way to test to make sure that you have successfully removed any of the memory leaks is to use the **`valgrind`** command.

Since this project makes extensive use of dynamic memory, it is important that you test your program for memory leaks using **`valgrind`**:

`valgrind ./proj3`

Note: If you <u>accidentally</u> use **`valgrind make run`**, you may end up with some memory that is still reachable. Do not test this – test using the command above where you include the input file. The **`makefile`** should include **`make val`** (which is ok).

If you have no memory leaks, you should see output like the following:

```
==5606==
==5606== HEAP SUMMARY:
==5606==     in use at exit: 0 bytes in 0 blocks
==5606==   total heap usage: 87 allocs, 87 frees, 10,684 bytes allocated
==5606==
==5606== All heap blocks were freed -- no leaks are possible
==5606==
==5606== For counts of detected and suppressed errors, rerun with: -v
==5606== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 6 from 6)
```

The important part is “in use at exit: 0 bytes 0 blocks,” which tells me all the dynamic memory was deleted before the program exited. If you see anything other than "0 bytes 0 blocks" there is probably an error in one of your destructors. We will evaluate this as part of the grading for this project.

Additional information on **`valgrind`** can be found here: http://valgrind.org/docs/manual/quick-start.html

Once you have compiled using your **`makefile`**, enter the command **`./proj3`** to run your program. You can use **`make val`** to test each of the input files using **`valgrind`** (do NOT use **`valgrind make run`**!). They have differing sizes. If your executable is not **`proj3`**, you will lose points. It should look like the sample output provided above.

7. Completing your Project

When you have completed your project, you can copy it into the submission folder. You can copy your files into the submission folder as many times as you like (before the due date). We will only grade what is in your submission folder.

For this project, you should submit these files to the **`proj3`** subdirectory:

`proj3.cpp` — should be unchanged.

`Node.h` — should be unchanged.

`Treatment.h` — should be unchanged.

`Treatment.cpp` — should include your implementation of the class functions.

`TreatmentCatalog.h` – should be unchanged.

`TreatmentCatalog.cpp` – should include your implementations of the class functions.

`PatientTreatment.h` — should be unchanged.

`PatientTreatment.cpp` – should include your implementations of the class functions.

`Patient.h` — should be unchanged.

`Patient.cpp` — should include your implementations of the class functions.

`TreatmentList.h` — should be unchanged.

`TreatmentList.cpp` — should include your implementations of the class functions.

`TriageSystem.h` — should be unchanged.

`TriageSystem.cpp` – should include your implementations of the class functions including the functions provided.

As you should have already set up your symbolic link for this class, you can just copy your files listed above to the submission folder. You should turn in all files for this project.

Step 1: cd to your project 3 folder. An example might be cd **`~/202/projects/proj3`**

Step 2: **`cp proj3.cpp TriageSystem.h TriageSystem.cpp Patient.h Patient.cpp TreatmentList.h TreatmentList.cpp PatientTreatment.h PatientTreatment.cpp TreatmentCatalog.h TreatmentCatalog.cpp Treatment.h Treatment.cpp Node.h~/cs202proj/proj3`**

You can check to make sure that your files were successfully copied over to the submission directory by entering the command

`ls ~/cs202proj/proj3`

You can check that your program compiles and runs in the **`proj3`** directory, but please clean up any **`.o`** and executable files. Again, do not develop your code in this directory and you should not have the only copy of your program here. Uploading of any **`.gch or vgcore*`** files will result in a severe penalty.

For additional information about project submissions, there is a more complete document available in Blackboard under “Course Materials” and “Project Submission.”

IMPORTANT: If you want to submit the project late (after the due date), you will need to copy your files to the appropriate late folder. If you can no longer copy the files into the proj3 folder, it is because the due date has passed. You should be able to see your proj3 files, but you can no longer edit or copy the files in to your proj3 folder. (They will be read only)

- If it is 0-24 hours late, copy your files to **`~/cs202proj/proj3-late1`**
- If it is 24-48 hours late, copy your files to **`~/cs202proj/proj3-late2`**
- If it is after 48 hours late, it is too late to be submitted.